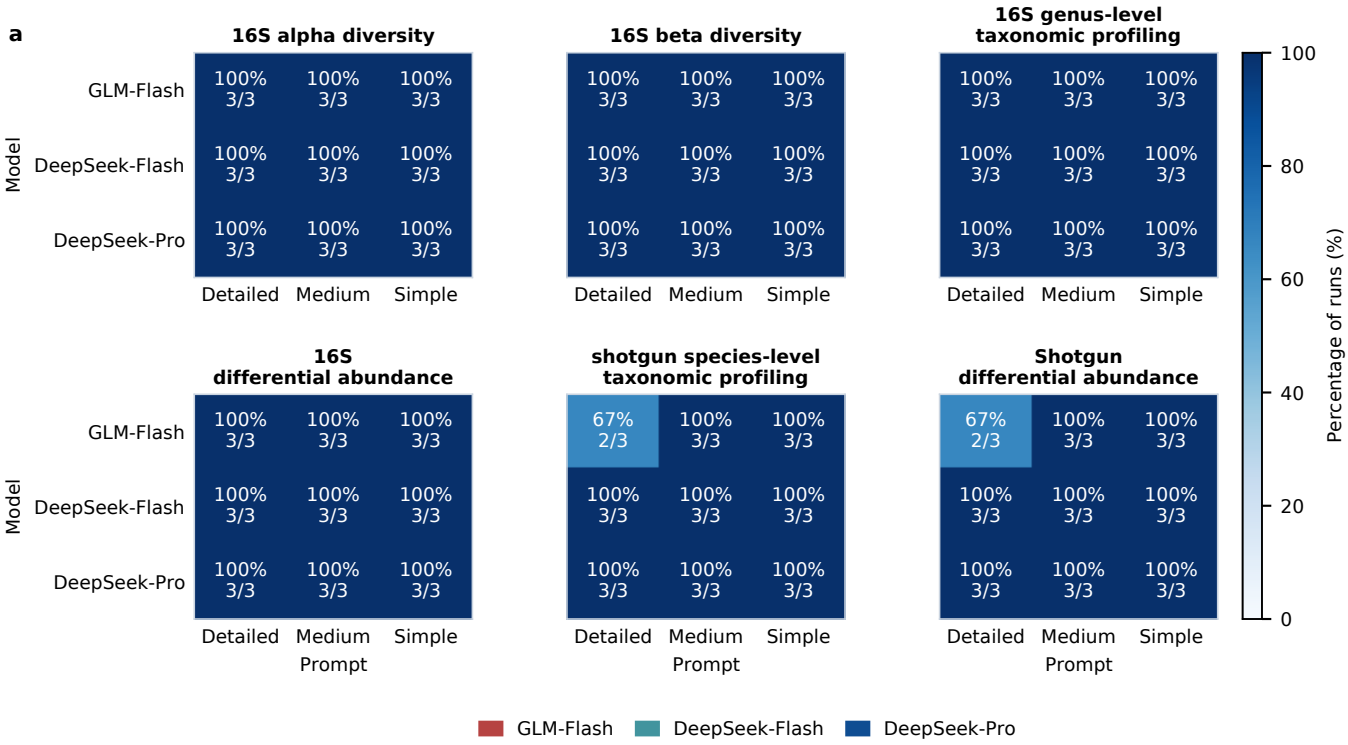
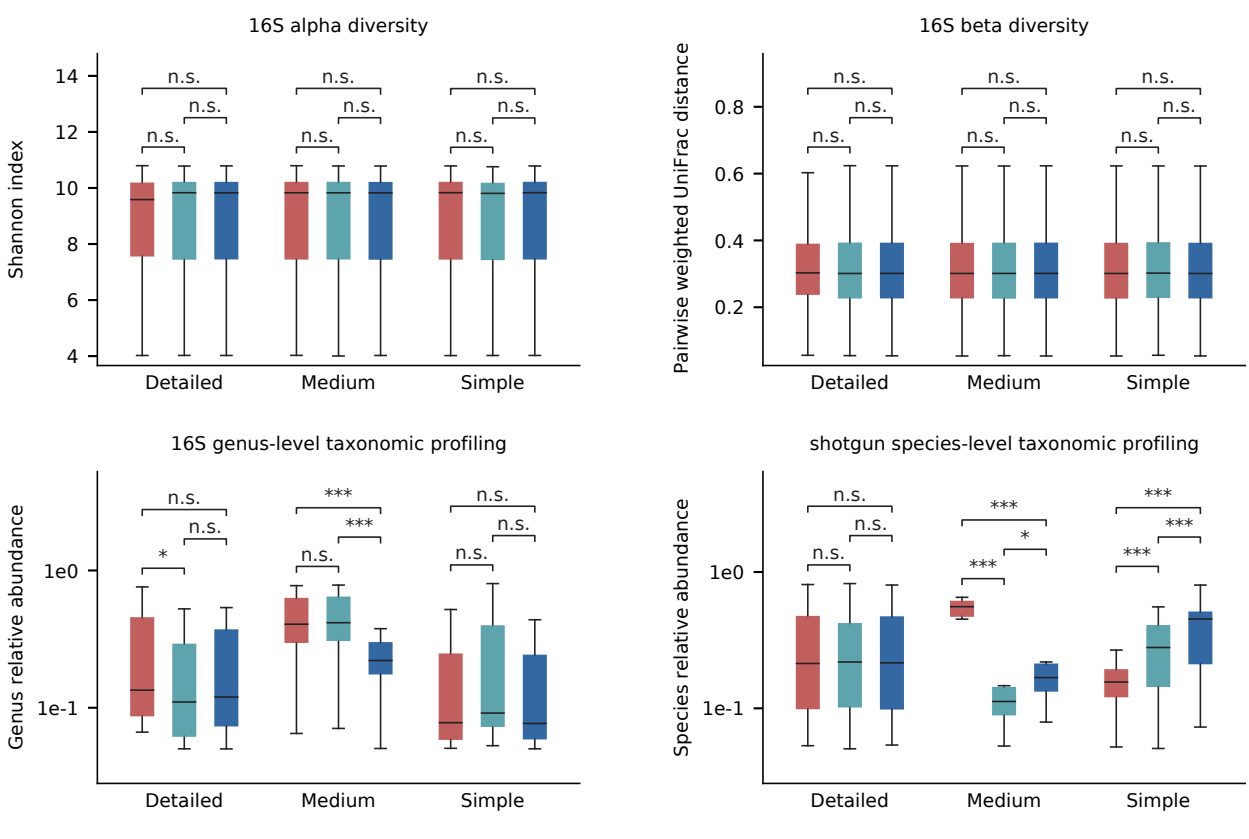
a

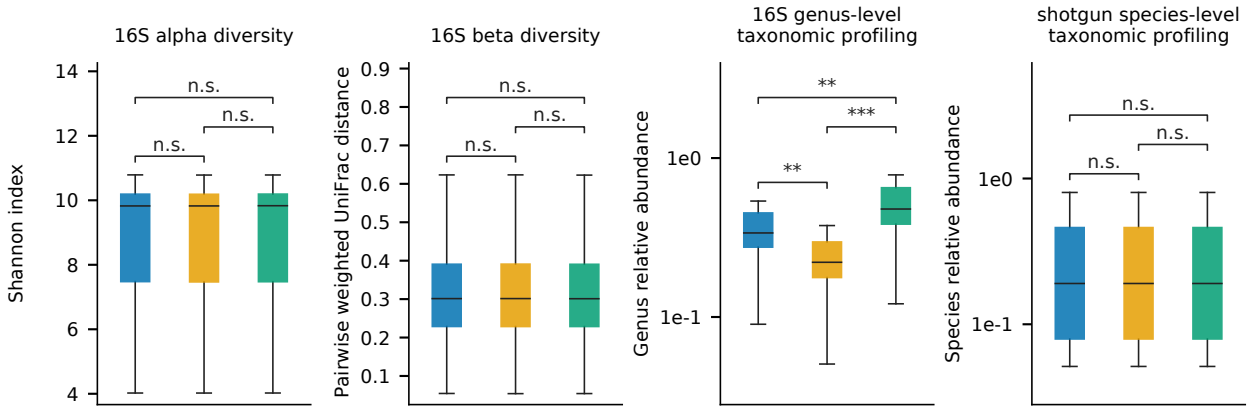

16S alpha diversity
16S beta diversity
16S genus-level taxonomic profiling
16S differential abundance
shotgun species-level taxonomic profiling
Shotgun differential abundance
Model
GLM-Flash
DeepSeek-Flash
DeepSeek-Pro
100% 3/3
67% 2/3
Detailed
Medium
Simple
Prompt
Percentage of runs (%)
0
20
40
60
80
100

GLM-Flash
DeepSeek-Flash
DeepSeek-Pro

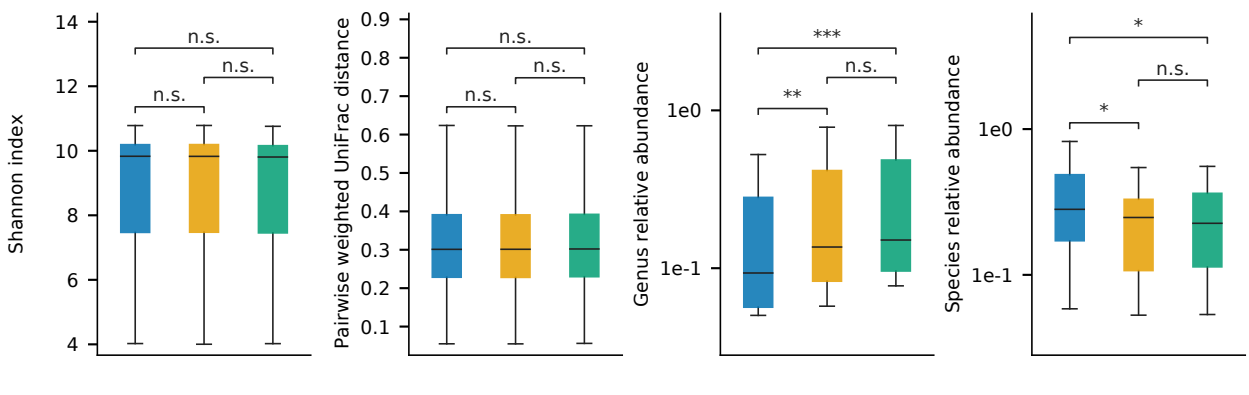
b


16S alpha diversity
Shannon index
16S beta diversity
Pairwise weighted UniFrac distance
16S genus-level taxonomic profiling
Genus relative abundance
shotgun species-level taxonomic profiling
Species relative abundance
n.s.
*
**
***
Detailed
Medium
Simple

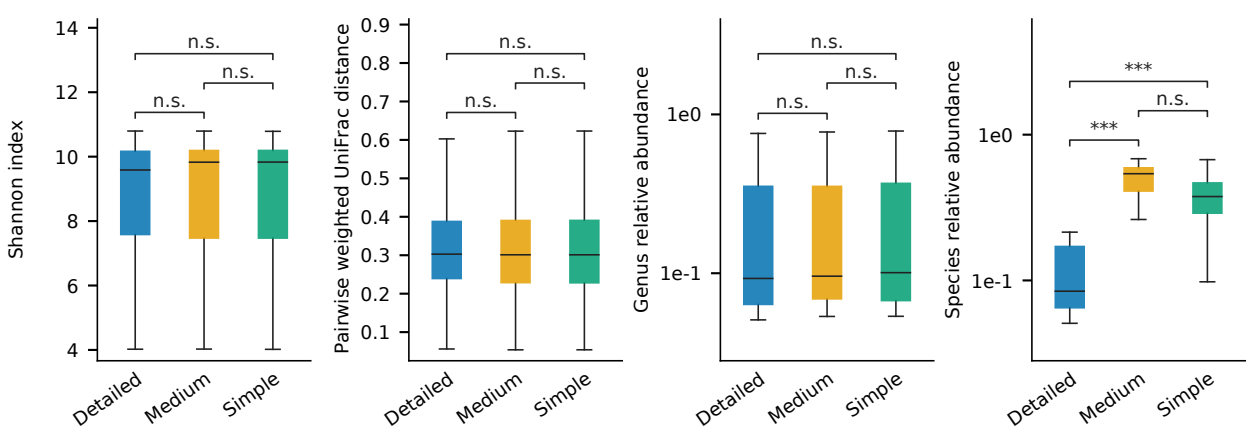
c


DeepSeek-Pro
16S alpha diversity
Shannon index
16S beta diversity
Pairwise weighted UniFrac distance
16S genus-level taxonomic profiling
Genus relative abundance
shotgun species-level taxonomic profiling
Species relative abundance
n.s.
*
**
***
Detailed
Medium
Simple

d


DeepSeek-Flash
Shannon index
Pairwise weighted UniFrac distance
Genus relative abundance
Species relative abundance
n.s.
*
**
***
Detailed
Medium
Simple

e


GLM-Flash
Shannon index
Pairwise weighted UniFrac distance
Genus relative abundance
Species relative abundance
n.s.
*
***
Detailed
Medium
Simple